Run report: sessions A and B (stride targets), 2026-10-14 17:00Z

Study: `research/ecbench_rootfind_speedup_20261014` (Conductor T-1). **Protocol:** `PROTOCOL.md`, committed in `07dd2d044` before these runs. **Binary:** `ecbench` SHA-256 `62bb5363f4dfca85…` (built at `07dd2d044`). **Host:** 4 vCPU Intel Xeon 2.10 GHz, shared cloud container; `--cpus none --allow-busy`, isolation **L0** on every record. **Runner:** `run.sh`.

session	spec	curve	targets × rounds	records
A `ECBS1h66ed5f6ccf76`	`claim-koblitz-rootfind-stride-n13.json`	`icv1-f2m13-t181-515ee569`, r = 2003	32 × 3 (+1 warm-up)	768 / 768 verified
B `ECBS1ha187c9428699`	`check-koblitz-rootfind-stride-n19.json`	`icv1-f2m19-t797-b6cf2467`, r = 130873	16 × 2 (+1 warm-up)	144 / 144 verified

Audits (`verify --replay-all`): A ok, 768 records; B ok, 144 records (`audits/audit-*.json`); both re-audited ok by the Amendment 1 binary (`audits/reaudit-*.json`). No run failed or timed out.

Measured S (GAE / √r), mean over measured runs, 95 % interval as `ecbench table` recorded

arm	A (n = 13)	B (n = 19)	priced
rho-strong	15.260 [14.156, 18.247]	3.112 [3.023, 3.205]	lower bound
rho-frob	1.870 [1.774, 2.060]	—	lower bound
d5-walk / d6-walk	22.996 [22.186, 23.900]	18.353 [16.517, 20.286]	A full; B lower bound
d5-stride / d6-stride	13.436 [12.814, 14.326]	21.036 [18.201, 23.838]	A full; B lower bound
d4-stride	11.375 [10.522, 12.298]	—	full
rho-strong-aa	15.260 [14.156, 18.247]	—	lower bound

Paired comparisons (`ecbench compare`, candidate ÷ baseline, cluster interval)

- A: d5-stride ÷ d5-walk = 0.5843 [0.548, 0.629], 96 pairs on 32 workloads.
- A: d4-stride ÷ rho-strong = 0.7454 [0.618, 0.845] (diagnostic only: rho-strong has unpriced work).
- A: d5-stride ÷ rho-strong = 0.8805 [0.722, 0.991] (diagnostic only).
- A: rho-strong-aa ÷ rho-strong = 1.0000 [1.000, 1.000].
- B: d6-stride ÷ d6-walk = 1.1462 [0.960, 1.363], 32 pairs on 16 workloads (diagnostic only).
- B: d6-stride ÷ rho-strong = 6.7601 [5.853, 7.648]; d6-walk ÷ rho-strong = 5.8979 [5.242, 6.568].

Recorded counters (means over measured runs): A d5-walk 118.9 trials, rank 9.17, rows 9.18; A d5-stride 200.6 trials, rank 12.14, rows 14.94; B d6-walk 3,998.7 trials, rows = rank 16.28; B d6-stride 6,347.5 trials, rank 23.41, rows 29.56. Wall medians are L0 and descriptive only.

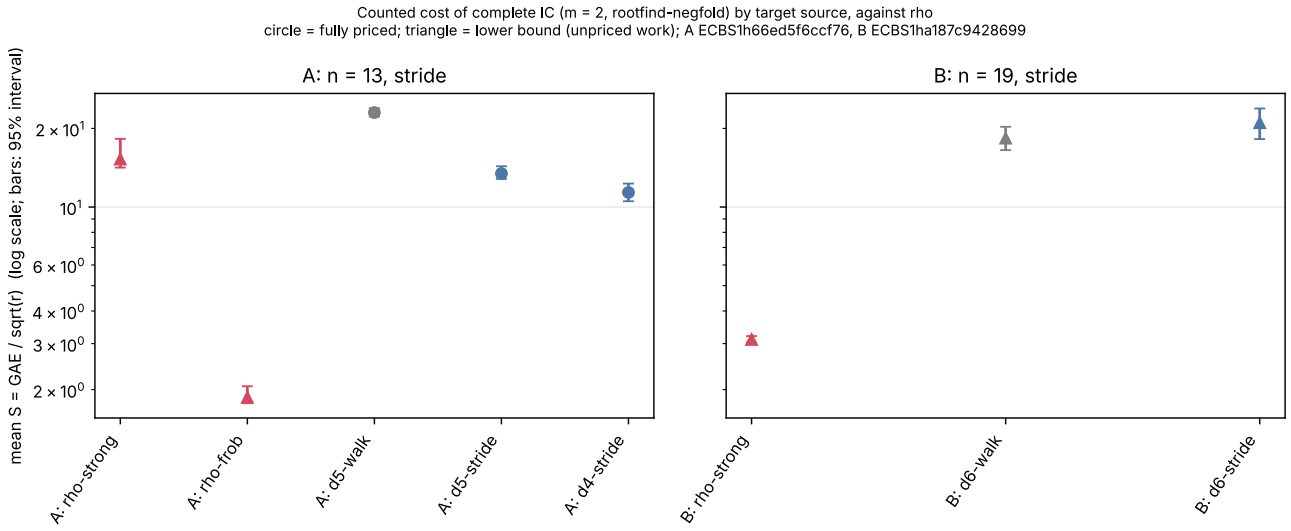


Figure 1: Mean S with the recorded 95 % interval per arm, sessions A (`ECBS1h66ed5f6ccf76`) and B (`ECBS1ha187c9428699`); circles fully priced, triangles lower bounds.

Raw records: `sessions/claim-n13/records.jsonl`, `sessions/check-n19/records.jsonl`; tables and comparisons under `tables/`.